

Experiment 1: Charpy V-notch test

Objectives

Understanding the behavior of brittle and ductile materials under high strain rate using the Charpy V-notch test.

Background

We have all seen solid material breaking around us in one form or another. Some materials break without undergoing significant deformations (e.g. a brick, a biscuit, glass), while others deform significantly before rupture commences (e.g. aluminum, lead, copper, etc.).

In this experiment, we will quantify this behavior of different materials under high strain rate (or impact loading). To that end, answer the following questions:

- 1.) Read, from a resource of your choice, the definitions of brittle and ductile materials.
Explain the difference between the two.
- 2.) How is the ductility (or brittleness) of a material dependent on its temperature?
- 3.) Does the ductility (or brittleness) of a material depend on the strain rate with which it is loaded? If yes, then describe how.
- 4.) Define *toughness* of a material.
- 5.) How is *toughness* of a material different from its *strength*.

Some online resources

Watch the following videos to learn how to perform a Charpy V-notch test. Small details may vary from one machine to another, but the essential objectives remain the same.

[Charpy Test Essentials](#) (36:54 min)

[Charpy Test-I](#) (4:56 min)

[Charpy Test-II](#) (4:31 min)

The experiment

Task 1:

A Charpy V-notch test machine is available in the Mechanical Lab. You will be given 2-3 samples of different materials, and you will be required to use the Charpy test machine to determine their toughness experimentally. Use the three links posted in the previous section to learn how to perform the experiment. Carry out the experiment and use it to estimate the toughness of the materials you test. Seek help from the staff members and the TAs if you need.

Task 2:

Write a report on the entire exercise. **The report must not exceed 3 pages.** You can tailor your report as per the following proposed structure, but you are free to deviate from it if you wish so.

- 1.) **INTRODUCTION:** State your objectives, and the answers to questions 1-5 stated in the “Background” section above.
- 2.) **PROCEDURE:** Describe the details of the procedure used to conduct the experiment.
- 3.) **OBSERVATIONS:** Report all the experimental data recorded. Describe your observations, insights, any calculations you perform.
- 4.) **CONCLUSIONS:** Describe what you understood from this entire exercise.